Task 3: Modelling Budgets and Downside Risk

Table of contents

1 Task 3: Modelling Budgets and Downside Risk **1**
- 1.1 Introduction . . . 1
- 1.2 Universal Pictures Budget Modelling . . . 1
- 1.3 Universal Pictures Budget Modelling . . . 1
- 1.4 Model Diagnostics and Comparison . . . 2
- 1.5 Action Genre Downside Risk . . . 3
- 1.6 Comparison of Interval Estimates . . . 3
- 1.7 Summary for the Finance Team . . . 3

1 Task 3: Modelling Budgets and Downside Risk

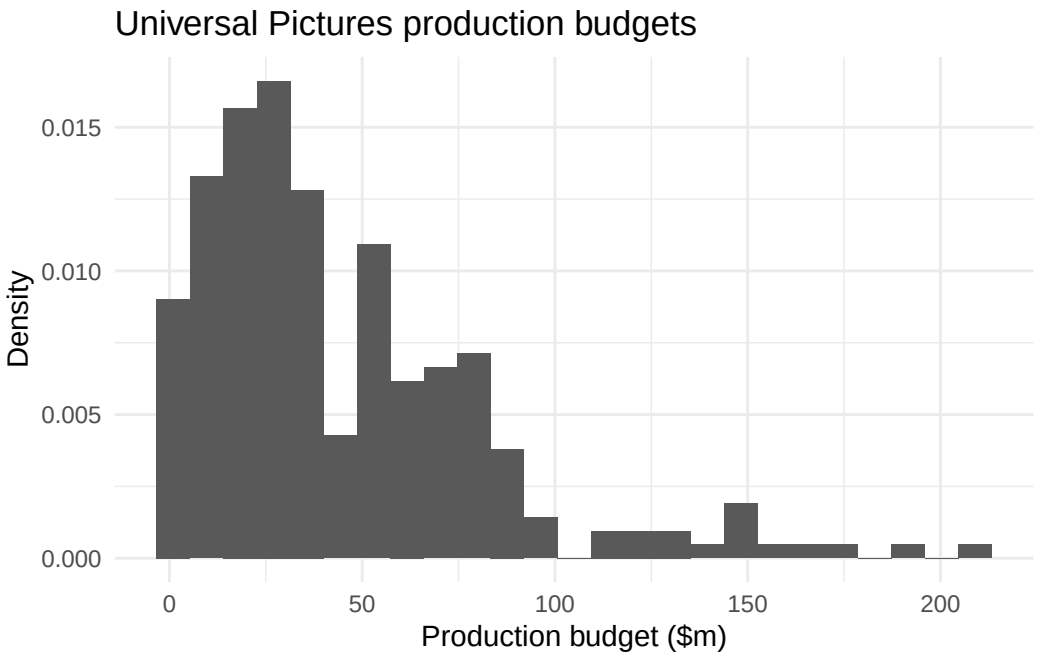
1.1 Introduction

[Explain the two financial planning questions.]

1.2 Universal Pictures Budget Modelling

1.3 Universal Pictures Budget Modelling

[Document the distributions considered.]



1.4 Model Diagnostics and Comparison

Distribution	Parameters (units)	Estimate 1	Estimate 2	Log-likelihood	AIC
Gamma	shape, rate (per $m)	1.382	0.031	-1,158.1	2,320.1
Weibull	shape, scale ($m)	1.214	47.479	-1,158.3	2,320.6
Log-normal	meanlog, sdlog (log $m)	3.392	1.015	-1,172.6	2,349.2
Normal	mean, sd ($m)	44.481	37.432	-1,225.1	2,454.1

Table 1: Distributions fitted to Universal Pictures budgets, ordered by AIC (lowest first)

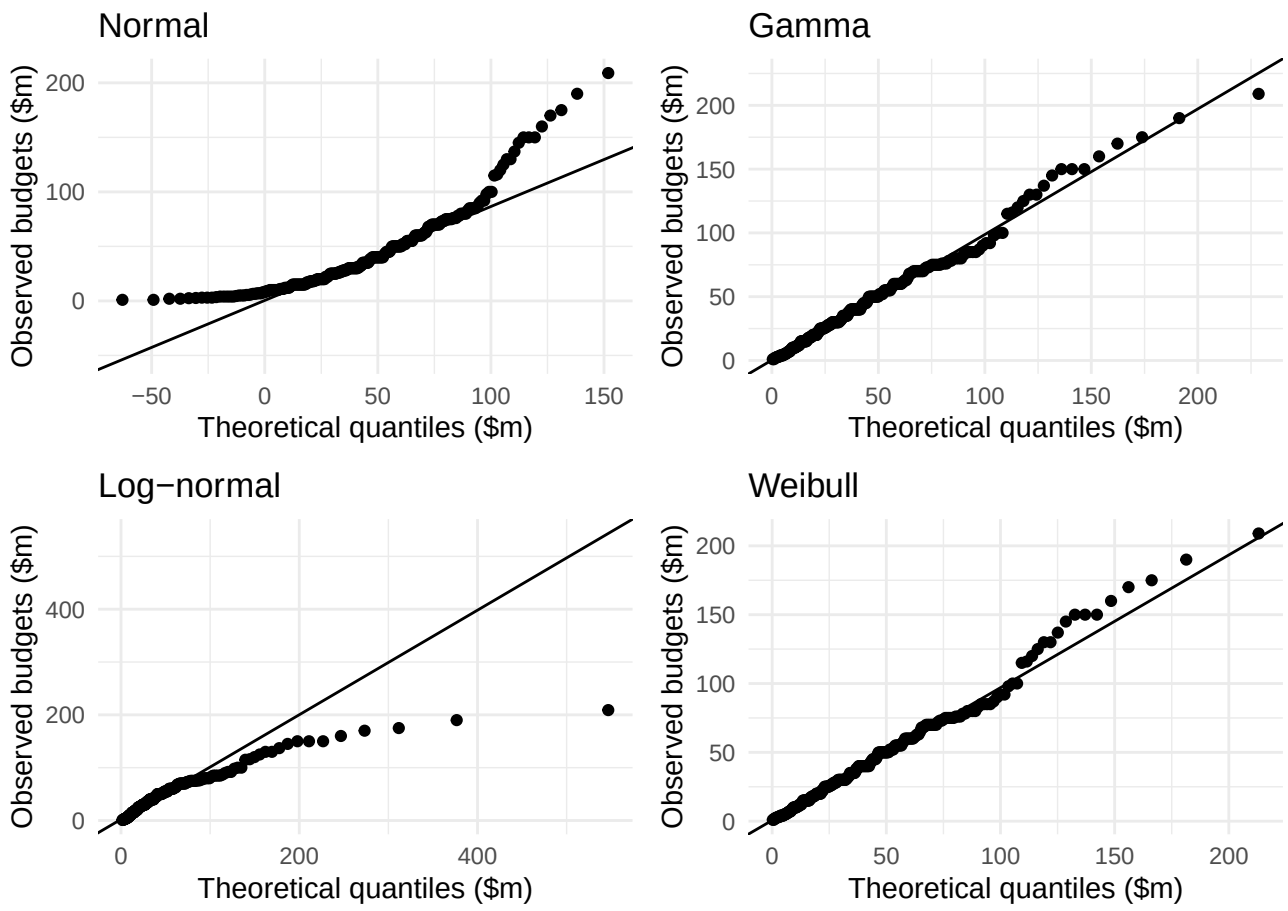


Figure 1: Q-Q plots of Universal Pictures budgets against each fitted distribution. Points close to the line indicate a good fit.

1.5 Action Genre Downside Risk

1.6 Comparison of Interval Estimates

[Compare methods, assumptions and findings.]

1.7 Summary for the Finance Team

[Summarise the implications in plain language.]